Data Management Project

Market Intelligence for Used-Car Export: Integrated Analysis of the European Market and U.S. Consumer Preferences

Nicolò Bachiorri[a,1], Davide Francesco Caramia[a,2] and Hervé Mottaran[a,3]

[a]*Università degli Studi di Milano Bicocca*

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Abstract—The European used-car market offers diverse vehicles with technical specifications and equipment configurations that could meet U.S. buyer preferences. This study develops a data integration framework combining EU supply data from AutoScout24 with U.S. consumer ratings from Cars.com. Through web scraping, normalization, and machine learning classification (Gradient Boosting), we identify which EU listings align with U.S. preferences. The integrated dataset (1507 rows, 239 unique make/model pairs) is stored in SQLite for reproducible querying. Model performance achieves ROC-AUC of 0.88, with interpretability analysis revealing that body type, engine power, and engine size are key factors influencing U.S. consumer appreciation.

1. WORK ORGANIZATION

The group consists of 3 members: N. Bachiorri, D.F. Caramia, H. Mottaran. Project work is organized around three main notebooks (EU scraping, US scraping, merging analysis) and a shared artifacts folder (`data`, `logs`, `checkpoints`). This structure supports reproducibility and clear traceability of intermediate outputs.

2. GOAL OF THE STUDY

2.1. Project Overview

The European used-car market offers a wide availability of vehicles with diverse technical specifications and equipment configurations that could potentially meet the preferences of U.S. buyers. However, assessing which models and configurations are likely to resonate with U.S. consumer preferences requires integrating EU supply data with signals from the U.S. market. This study develops a data integration framework to address this gap, enabling preference-driven selection of EU used-car listings based on U.S. consumer ratings.

2.2. Research Questions

1. **Market matching:** Which used-car models offered in the European market are also present in the U.S. market and can therefore be assessed against U.S. consumer preferences?
2. **Preference-driven selection:** Among EU listings that match U.S. models, which technical characteristics and equipment options are associated with higher U.S. consumer preference signals?
3. **Decision support:** How can the integrated dataset support investment decisions by ranking EU listings according to expected U.S. market appeal?

2.3. Steps of the Study

- Web data acquisition from AutoScout24 (EU supply) and Cars.com (US preferences).
- Normalization of matching keys (make/model) and harmonization of schemas.
- Integration into a unified dataset and storage in SQLite for reproducible querying.
- Data quality assessment (completeness and uniqueness).
- Target definition via weighted PCA and supervised classification (Gradient Boosting).
- Interpretability analysis (native feature importance and LIME).

This report was prepared for the Data Management course at Università degli Studi di Milano Bicocca. The project integrates European used-car supply data with U.S. consumer preferences to support investment decisions in the automotive export market.

3. DATASET (ACQUISITION & PREPARATION)

3.1. AutoScout24 (European Market Supply)

Acquisition was performed in Python using `requests` and `BeautifulSoup`, adopting a "polite scraping" strategy (realistic headers, delays between requests, checkpoints and logs). In the snapshot collected on December 29-2025, 3633 listings were retrieved (after removing 366 duplicates), and detailed technical information was successfully parsed for 3532 vehicles.

3.1.1. *Main Artifacts (EU)*

- `data/autoscout24_listings.json` (raw listings, unique IDs; duplicates removed during collection).
- `data/autoscout24_car_details.json` (raw detail pages parsed into structured fields).
- `data/autoscout24_combined_data.json` (flattened dataset with `spec_*` and `feat_*` fields).
- `logs/scraping_log_*.txt` and `logs/car_details_log_*.txt` (trace of pages/vehicles processed, pauses, failures).
- `checkpoints/scraping_checkpoint.json` and `checkpoints/car_details_checkpoint.json` (resume capability).

3.1.2. *Scraping Reliability and Politeness (EU)*

- Rotating User-Agent list and realistic HTTP headers for each session.
- Timeouts on HTTP requests and exception handling to avoid interrupting long runs.
- Long pauses between listing pages (randomized) and short human-like pauses between detail pages.
- Automatic checkpoints (e.g., every 10 pages) and resumable execution.

3.2. Cars.com (U.S. Consumer Preferences)

Acquisition was performed with Selenium to navigate Cars.com research pages and extract multi-dimensional ratings at model level. The scraping workflow includes explicit politeness delays between requests and complete execution logs.

3.2.1. *Main Artifacts (US)*

- `data/carscom_enriched.json` (structured extraction and aggregation metadata).
- `data/carscom_flat_weighted.json` (flat table with rating dimensions and review volume – `cumulata`).
- `logs/carscom_log_*.txt` (run trace).
- `checkpoints/cars_reviews_checkpoint.json` / `carscom_checkpoint.json` (resumable execution).

3.3. Snapshot Summary

Table 1. Dataset components and key metrics

Component	Size / key numbers
AutoScout24 listings	3633 listings (duplicates removed: 366); details for 3532 vehicles; date 2025-12-29
Integration result	1507 rows; 239 unique make/model pairs with `cumulata` >0
SQLite storage	`my_project.db` – table `cars_data` (1507 rows, 174 columns)
Final ML dataset	1486 rows; `appreciate=False` 936, `appreciate=True` 550

4. STRUCTURE ON SQLITE (DBMS)

4.1. Storage Choice

The integrated dataset is persisted in SQLite to enable fast inspection through SQL queries and to provide a single, portable artifact of the curated dataset. The project database is stored as `my_project.db` and contains a single table, `cars_data`, with 1507 rows and 174 columns.

4.2. Table Content (Logical Grouping)

- Identifiers and listing metadata (e.g., `id`, `make`, `model`, `year`, `price`, `mileage`).
- Technical specifications from AutoScout24 (`spec_*` fields).
- Equipment features from AutoScout24 (`feat_*` binary fields).
- U.S. preference signals from Cars.com (`comfort`, `interior`, `performance`, `value`, `exterior`, `reliability`) and review volume (`cumulata`).

4.3. Example SQL Queries

Two queries used for basic exploration and sanity checks are shown in *query1* and *query2* (output truncated).

```sql
SELECT DISTINCT
    make,
    model
FROM cars_data
WHERE spec_fuel_consumption < 7.0;
```

Index	Make	Model
0	bmw	320
1	ford	focus
2	ford	fiesta
3	fiat	500
4	jaguar	e-pace
5	audi	a3
6	jaguar	xe
7	volkswagen	tiguan
8	bmw	x3
9	toyota	yaris
10	volkswagen	golf
11	bmw	x1
12	jaguar	f-pace
13	bmw	318
14	audi	a4
15	audi	q3
16	land rover	range rover evoque
17	volvo	v90
18	bmw	530
19	audi	a4 allroad
20	land rover	range rover sport

```sql
SELECT
    make,
    AVG(price)
FROM cars_data
GROUP BY make
ORDER BY AVG(price) DESC;
```

Index	Make	AVG(price)
0	rolls-royce	399640.00
1	lamborghini	276130.00
2	ferrari	214830.00
3	mclaren	149570.00
4	aston martin	99999.00
5	ram	89500.00
6	maserati	75797.14
7	bentley	72983.33
8	chrysler	72750.00
9	porsche	66893.70
10	mercedes-benz	53997.24
11	land rover	47555.70
12	bmw	42855.08
13	dodge	41119.67
14	audi	32548.89
15	lexus	30566.17
16	tesla	29548.59
17	lotus	28900.00
18	alfa romeo	28543.68
19	volvo	25812.61
20	polestar	24946.67

5. DATA INTEGRATION

5.1. Integration Strategy

Integration is implemented by normalizing `make` and `model` into a slug form (lowercase, underscores/alphanumeric) and performing an inner join between EU listings and U.S. ratings. This choice ensures that only EU vehicles with measurable U.S. preference data are retained. The join key is the tuple `(make_slug, model_slug)`.

5.2. Integration Output

The integrated dataset contains 1507 rows. After filtering entries with `cumulata = 0` (no U.S. reviews), the number of unique make/-model pairs used for modeling is 239. Main integration artifact: `data/combined_listings_reviews.csv` (EU listings enriched with U.S. ratings at make/model level).

6. DATA ENRICHMENT

6.1. Numeric Normalization of Selected Attributes

Before modeling, selected text-formatted attributes are converted into numeric values to enable learning on comparable scales:

- `spec_empty_weight`: remove thousand separators and “kg” suffix, cast to float.
- `power`: parse the kW value from strings like “135 kW (184 hp)”, cast to numeric (invalid values coerced to NaN).
- `spec_engine_size`: remove thousand separators and “cc” suffix, cast to float.
- `spec_fuel_consumption`: extract the first numeric token and cast to float.

6.2. Target Definition (Preference Proxy)

The target variable `appreciate` (binary) is derived from a weighted PCA applied to the six rating dimensions (`comfort`, `interior`, `performance`, `value`, `exterior`, `reliability`), using `cumulata` (review volume) as sample weights. The first principal component (PC1)

is used as a composite preference score, and the binary target is defined as:

$$\text{appreciate} = (\text{PC1} > \text{mean}(\text{PC1})) \quad (1)$$

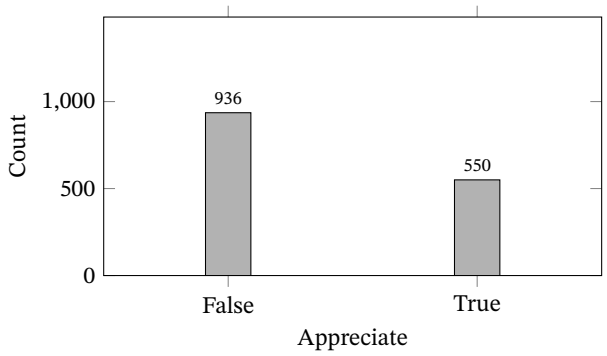


Figure 1. Class distribution in the final ML dataset (1486 rows)

7. ANALYSIS

7.1. Data Quality Assessment

We performed a basic data quality assessment of the integrated dataset along two measurable dimensions: completeness and uniqueness.

7.1.1. Completeness

Completeness is computed as $1 - \frac{\text{Null}}{m \times n}$, where Null is the number of missing values and $m \times n$ is the total number of cells in the dataset. **Measured completeness:** 98.85%.

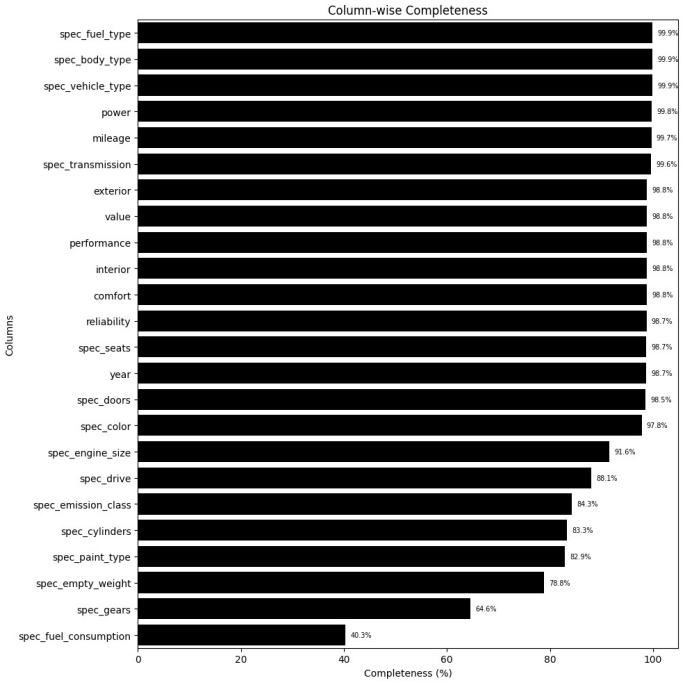


Figure 2. Column completeness (%) in the integrated dataset (only columns with < 100% completeness).

7.1.2. Uniqueness

Uniqueness is computed as $\frac{\#\text{unique ids}}{m}$. It measures the fraction of distinct identifiers over the number of rows. **Measured uniqueness:** 99.8% (with 3 duplicated ids).

7.2. Machine Learning Classification

A `GradientBoostingClassifier` is trained to predict `appreciate`. The dataset is split into train/test with `test_size=0.2`, `random_state=42`, and stratification on the target. Before training, object-type features are one-hot encoded via `pd.get_dummies()`.

7.2.1. Model Performance (Test Set)

GridSearchCV configuration: `cv=5`, `scoring='roc_auc'` (reported CV ROC-AUC: 0.8624).

Table 2. Model performance metrics

Metric	Baseline	Tuned
Accuracy	0.8020	0.8322
Precision	0.7576	0.7830
Recall	0.6818	0.7545
F1	0.7177	0.7685
ROC-AUC	0.8571	0.8838

7.3. Interpretability: Feature Importance and LIME

To improve interpretability, we compare (i) the native feature importance provided by Gradient Boosting and (ii) LIME-based explanations aggregated globally. Unlike native importance, LIME also provides directional information (positive or negative contribution to the target class). Fig. 3 shows the comparison between native and LIME-based feature importance.

The native model highlights `power`, `spec_engine_size`, and `spec_body_type_Off-Road/Pick-up` among the most relevant variables. The aggregated LIME analysis emphasizes `spec_body_type_Off-Road/Pick-up` and `power` but also reveals directional patterns not directly visible in the native importance ranking.

7.3.1. Key Directional Insights

- `spec_body_type_Convertible` shows the strongest positive directional contribution, significantly increasing the probability of appreciation.
- `power` and `spec_engine_size` contribute positively and are consistent with a preference for higher-performance vehicles.
- `spec_fuel_type_Electric` shows the strongest negative contribution, reducing the probability of appreciation in the learned preference proxy.
- Negative contributions are also observed for `spec_body_type_Van` and `spec_body_type_Other`, as well as for `feat_right_hand_drive` and some comfort-related features (e.g., `feat_automatic_climate_control`).

7.4. Software Architecture and Reproducibility

The project is organized as a notebook-driven pipeline with intermediate artifacts stored in the `data` folder and operational traceability ensured by `logs` and `checkpoints`.

- `AutoScout_Data_Collection.ipynb` → `autoscout24_listings.json`, `autoscout24_car_details.json`, `autoscout24_combined_data.json`
- `Reviews_Data_Collection.ipynb` → `carscom_enriched.json`, `carscom_flat_weighted.json`
- `Merging_Collections.ipynb` → `combined_listings_reviews.csv` → SQLite (`my_project.db`) → `combined_listings_reviews_dummified.csv` → `combined_listings_reviews_final.csv`
- `logs/` and `checkpoints/` provide traceability and resume capability for long-running collection tasks.

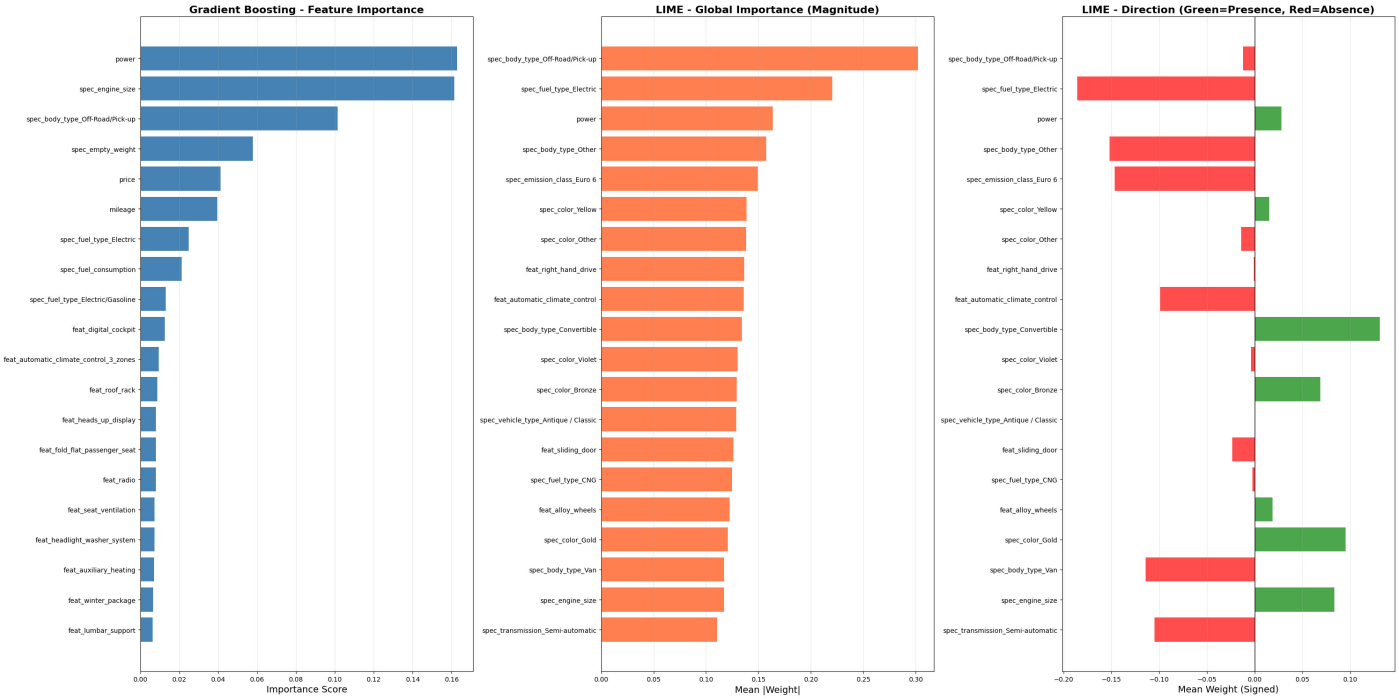


Figure 3. Feature importance comparison: Gradient Boosting native importance vs LIME global importance and LIME directional effects.

8. CONCLUSIONS

This project builds an integrated dataset connecting EU used-car supply (technical specifications and equipment) with U.S. consumer preference signals (ratings), enabling a preference-driven selection framework. A machine learning model trained on this integrated dataset achieves strong predictive performance (ROC-AUC = 0.88) and interpretability analysis reveals that body type (particularly Convertible and Off-Road/Pick-up), engine power, and engine size are among the most influential factors associated with U.S. consumer appreciation.

8.1. Limitations and Future Developments

- **Regulatory feasibility is not modeled:** exporting EU vehicles to the U.S. requires compliance with FMVSS/EPA regulations; feasibility and conversion costs are left to subsequent business evaluation.
- **Single time snapshot:** EU supply data were collected in one day; temporal dynamics (inventory turnover, price changes) are not captured.
- **Cross-market matching is constrained** by make/model alignment and by review availability on the U.S. side (`cumulata`).